

VAPOR PRESSURE MEASUREMENTS OF POTASSIUM OVER K_2O-SiO_2 SOLUTIONS BY A
KNUDSEN EFFUSION, MASS SPECTROMETRIC METHOD

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Measurements of the K pressure over K_2O-SiO_2 solutions were carried out using a modulated beam--quadrupole mass spectrometer. Pressure measurements were made as a function of composition and temperature and covered the composition range 43.9 to 11.9 wt percent K_2O and the temperature range 1300-1800 K. The composition variable was determined through use of the integrated ^{39}K ion current. In the above composition range, all of the data were well represented by a master equation with a single slope and a composition term in mole fraction squared. At still lower K_2O content, deviations from the master equation were apparent. The detailed data are presented.

1. Introduction

The concept of using MHD to obtain high efficiency combined-cycle energy conversion systems has led to a need for data which can be used to estimate the activity of K_2O dissolved in coal slag. Although K_2O is generally only a minor constituent of coal slags, the interaction of slags with the potassium seed required for MHD makes slags containing much larger concentrations of K_2O of interest. Since most coal slags contain of the order of 50 wt percent SiO_2 , solutions of K_2O in SiO_2 can serve as a simplified model reference slag.

There are few data in the literature relevant to the activity of K_2O in SiO_2 solutions. Preston and Turner [1]¹ measured rates of evaporation in air of various K_2O-SiO_2 melts as well as apparent pressures as a function of flow rate by the transpiration method. At the time their measurements were made, it was customary to obtain equilibrium pressures by extrapolation to zero flow rate, but this method has since been shown [2] to lead to pressures which are too high because of diffusion effects. Attempts to derive activity data

¹Figures in brackets indicate the literature references at the end of this paper.

based in whole or in part on Preston and Turner's rate of evaporation data have been made by Callow [3], Charles [4], and Eliezer, et al. [5]. Charles, and Eliezer, et al., recognized that the enthalpy change based on the rate of vaporization data was not consistent with evaporation of K and O_2 from the melts and concluded that the measurements would be better interpreted in terms of reaction with water vapor to form KOH(g).

While MHD applications require data for very complicated slag solutions which may contain up to six components, the use of a simplified two component solution as a standard may be helpful. The major objective of this paper will therefore be to describe the measurement technique and present the basic data obtained in studies on the evaporation of K and O_2 from K_2O-SiO_2 solutions.

2. Description of Mass Spectrometric System

The mass spectrometric system was the same as that previously used and described by Paule [6], but additional description of the system will be made to be in keeping with the emphasis of this volume on methodology.

The quadrupole mass spectrometer was constructed from components manufactured by Extranuclear Corp., Pittsburgh, Pa. The beam modulation system was constructed at NBS from commercially available components and consisted of a Wavetek model 130 function generator, a Bogen model Mo100A amplifier and a Globe Industries model 18A108 synchronous motor equipped with high vacuum bearings, which rotated a 10 toothed wheel to alternately pass or scatter the molecular beam effusing into the ion source from the Knudsen cell. A reference signal dependent on wheel position was generated by an HEI optical switch and fed to the reference circuit of an Ithaco model 391A lock-in amplifier. A small fraction of the neutral vapor species was ionized, focussed by the ion optics, and either transmitted or rejected by the quadrupole lens assembly depending on the mass selected. Molecules originating from the Knudsen cell traversed the mechanical chopping disc and bore a definite phase relationship to the reference signal. The phase angle for maximizing the amplitude was determined from an approximate phase angle by offsetting the selected phase angle by 90° , adjusting the phase control for 0 amplitude, and offsetting again by 90° . This procedure is used because the change in amplitude with phase angle is greatest at $\phi = 0$.

The distance from the Knudsen cell to the modulation disc was about 5 cm and from the modulation disc to the ion source was 10 cm. Although this system is theoretically capable of using phase sensitive detection to determine whether ion species are formed by direct ionization or fragmentation of a greater molecular weight precursor, such measurements were not necessary in this work and no attempt was made to use this capability of the modulation system. During the measurements, a constant modulation frequency of about 600 Hz was used.

The vacuum system was constructed from commercially available stainless steel components. Vacua were maintained using two liquid nitrogen trapped, four-inch, oil diffusion pumps. Two 4 inch gate valves allowed the system to be opened to the atmosphere without shutting down the diffusion pumps. The system differed from most mass spectrometric systems in that it did not make use of differential pumping. It was assumed that this arrangement

could help to avoid modulation of signals caused by the flow of background gases from a relatively high pressure Knudsen chamber (10^{-7} Torr) to a low pressure mass spectrometric chamber (10^{-8} Torr). However, the differential pumping effect could not be entirely eliminated because the Knudsen cell and furnace assembly were contained in a water-cooled copper chamber which had a high gas conductivity to one pump and lower gas conductivity to the second. In spite of this, pressures as measured by ionization gages which had line of sight connection to the respective regions usually did not differ significantly during experiments and were in the 10^{-7} - 10^{-8} Torr range.

Figure 1 shows a schematic of the mass spectrometer system. The quadrupole mass filter (QMF) is mounted about 1-1/2 inches off center on a 6-inch flange in a horizontal position. The Knudsen cell and furnace are contained within a water-cooled copper chamber in order to eliminate effects of outgassing during the experiments. This appears to be a worthwhile technique because the relatively poor thermal conductivity of stainless steel makes external cooling relatively ineffective. The water-cooled motor (M) which drives the modulation disk (MD) and the optical switch (OS) are located below the mass filter. It was found that water cooling of the motor essentially eliminated outgassing during operation. The location of ion gauges (I.G.), and a shutter (S) for protecting the window (W) are also indicated. A prism (P) on the top flange deflected the radiant energy emitted from the Knudsen cell into an optical pyrometer for temperature measurement, while temperature stability was monitored by a thermocouple (T.C.) fed through the bottom flange. The water-cooled electrodes for heating the furnace are insulated from the flange using Teflon sleeves and Swagelock fittings (details not shown). All other vacuum feedthroughs are made using commercial fittings.

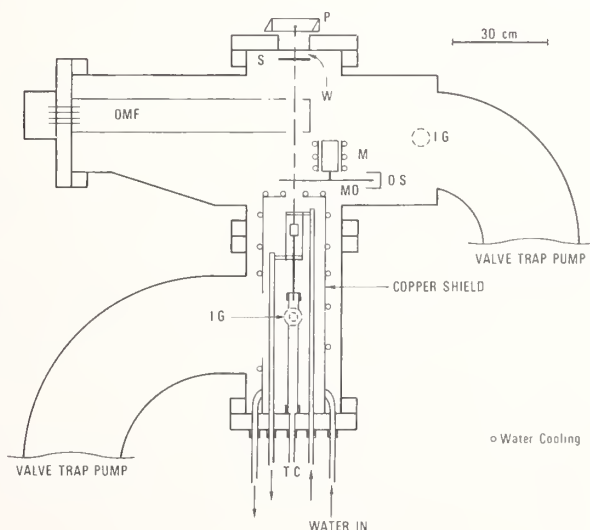


Figure 1. Schematic of vacuum enclosure and mass spectrometric system.

Figure 2 shows a schematic diagram of the Knudsen cell and furnace. The Knudsen cell was made of 0.025 cm Pt sheet and was of welded construction. It was 0.80 cm o.d. by 1.90 cm high and was welded to a 0.32 cm o.d. platinum mounting tube about 2.5 cm long. The top of the cell was made from 0.015 cm Pt sheet, the outer edge of which was bent in a mold to form a lip so that the lid fit snugly inside the effusion cell body (note detail in fig. 2). After loading the cell, the lid was spotwelded in numerous places around the lip to eliminate leakage.

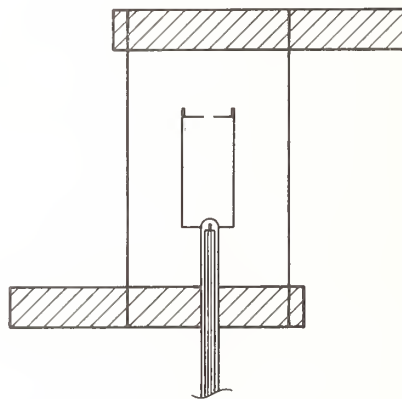


Figure 2. Schematic of furnace and Knudsen cell.

The furnace was made from 0.005 cm Ta sheet sandwiched between 0.62 x 1.6 diam x 0.05 cm Ta support rings at the ends and 0.62 x 0.05 cm Ta current leads which were connected to 1.25 cm o.d. water-cooled copper electrodes. In the first three series of measurements, the furnace was 5 cm long and the Knudsen cell was centrally located coaxially and vertically in the furnace.

After the first three series, the furnace was lengthened to 7.5 cm and the diameter of the furnace support rings was increased to 1.9 cm o.d. The latter change was made because most of the heat is absorbed by the lateral surface area of the effusion cell, and increasing the distance between the effusion cell and heater has the effect of smoothing temperature gradients in the effusion cell caused by temperature inhomogeneities in the heater or centering errors in the effusion cell. The cell position was biased upward by 0.6 cm to ensure that the base of the effusion cell was not in a cooler zone of the furnace and to compensate for heat losses due to the support. In all measurements, a single longitudinal radiation shield approximately 0.3 cm from the furnace and extending 2.5 cm below the furnace was used to increase the efficiency of the heater as well as to provide a more uniform hot zone.

Besides an increase in the furnace length, several changes in cell support were made as the measurements progressed. In the first series (1104), the Knudsen cell was mounted

on a 0.25 cm o.d. Mo rod which fitted snugly inside the Pt mounting tube. Because of the possibility that this could lead to undue cooling of the base of the effusion cell, an Al_2O_3 thermocouple insulation tube was installed and used as a support in subsequent work. A Pt/Pt, 10 percent Rh thermocouple was installed at the same time with the thermocouple hot junction in proximity but not touching the base of the cell in order to avoid sticking problems. Comparison between the optical pyrometer and thermocouple temperatures showed that the thermocouple systematically read low. This was probably due to poor thermal contact between the thermocouple junction and the cell.

With a furnace length of 7.5 cm starting with series 1122, agreement between the thermocouple temperature and the optical pyrometer temperature was more satisfactory than with the shorter furnace. The major use of the thermocouple was to indicate whether the cell temperature was constant. This was achieved by feeding the output of the thermocouple to a Doric Trendicator 400 Type S digital indicator which constantly displayed the thermocouple output in $^{\circ}\text{C}$. This allowed an almost instantaneous check to be made to determine if increases or decreases in the ion intensity were correlated with temperature changes of the effusion cell.

Power was controlled by a Stabiline constant voltage transformer. With all leads securely tightened on the low voltage-high amperage side of the power supply, temperature control was usually better than 2°C .

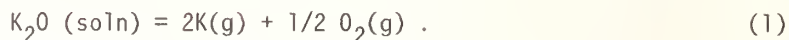
Temperature measurements were made using a Pyro-Micro-optical pyrometer, the temperature scale of which was verified by comparison with a bandlamp calibrated by the Temperature Physics Section at NBS. Temperatures were read through a prism, window, the ion source, and the modulation disc of the mass spectrometer, and into the black body orifice at the top of the effusion cell. Corrections of $-1.25 \times 10^{-5}\text{K}^{-1}$ for the window and prism, and $-3.13 \times 10^{-5}\text{K}^{-1}$ for the 50 percent transmission modulation disc were added to the reciprocal apparent absolute temperature to obtain a temperature on the IPTS-68 scale.

3. Procedure

Vapor pressure measurements over solutions would ideally be carried out by determining the pressure variation with composition at a number of isotherms or by determining the pressure variation with temperature at a number of constant compositions. Neither of these ideals was achieved, but at the lower temperatures, compositions are constant for all practical purposes and a large fraction of the change in composition takes place at the highest temperature points.

Because $\text{K}_2\text{Si}_2\text{O}_5$ is somewhat hygroscopic, the sample was dried overnight at 500°C prior to use and undue exposure to the atmosphere was minimized during loading of the effusion cell, cell alignment in the mass spectrometer, and weighing. In the initial measurements when the sample was first being heated, special care was taken to attempt to observe phase locked signals corresponding to H_2O , KOH which might be formed by reaction of $\text{K}_2\text{Si}_2\text{O}_5$ with absorbed water, and CO_2 which would be formed by residual, unreacted K_2CO_3 . However, no appreciable signals resulting from vaporization of these impurities were detected, indicating that sample purity was satisfactory.

The predominant vaporization process can be written as:



Throughout the range of these measurements, the loss of SiO(g) or K-bearing molecules was insignificant. After vaporization of sufficient sample to give an accurately measurable weight loss, the mass spectrometer calibration constant was determined by equating the pressure from the mass spectrometer equation:

$$P_i = k_i I_i^+ T \quad (2)$$

to the pressure calculated in a gravimetric Knudsen measurement,

$$P_i = \frac{m_i}{c a t} \sqrt{\frac{2\pi R T}{M_i}} \quad (3)$$

in which P_i is the pressure of some species i , k_i is the mass spectrometer constant, T the absolute temperature, m_i is the mass loss, c the Clausing factor, a the orifice area, t the elapsed time, R the gas constant, and M_i the molecular weight of species i . It should be noted that the mass spectrometer constant defined in eq. (2) implicitly assumes that the isotopic abundance A_i , is taken into account when only a single isotopic species is observed (in this case ^{39}K). Equating (2) to (3) and inserting the isotopic abundance factor gives,

$$k_i = [m_i A_i (2\pi R)^{1/2}] / [c a I_i^+ (M_i T)^{1/2}] . \quad (4)$$

In practice, the sum of weight losses for a number of experiments at different temperatures and for different periods of time are observed so that eq. (4) need be cast in the form,

$$k_i = [G A_i (2\pi R)^{1/2} \Sigma m_i] / [a c \sqrt{M_i} \Sigma I_i^+ t \sqrt{T}] . \quad (5)$$

For vaporization of K_2O from $\text{K}_2\text{O-SiO}_2$ solutions, Σm_i is the weight loss of K_2O and G is a gravimetric factor to convert the weight loss of K_2O to K . For each experimental point, the composition of the evaporating solution was determined from the known starting composition and the summation of $I_i^+ t \sqrt{T}$ terms. At the higher temperatures, there was a measurable decrease in the pressure (ten percent) and change in solution composition with time, but the change in pressure was essentially linear with time. The pressure and K_2O concentration calculated at the mean time interval were accepted for each observation in most cases, but in a few of the earlier runs, pressures were extrapolated to a known composition.

Each time the Knudsen cell was removed from the mass spectrometer, it was weighed so that the weight loss and $\Sigma I^+ t \sqrt{T}$ data could be used to calculate the mass spectrometer sensitivity constant. Three mass spectrometer constants and the series designation they were used for are listed in table 1. In the last series of data, 1214, a valid calibration constant was not obtained because deformation of the furnace in the latter part of the experiment caused excess weight loss due to vaporization of platinum from the crucible. The calibration constant used in this series was the average of the three previously determined values.

Table 1. Mass spectrometer constants.

Series Designation	k
	Microvolts atm ⁻¹ K ⁻¹
1104	1.96 E-11
1110-1115	1.47 E-11
1122-1129	1.83 E-11
1214	1.75 E-11 ^a

^aAverage of previous three values.

In spite of the modulated beam system, some extraneous phase locked mass 39 signals at low pressures ($P_k < 10^{-7}$ atm) were apparent. Because a rapid increase in this signal coincided with the increase in power to the furnace and slowly decayed away, it appeared likely that it was due to desorption of residual hydrocarbon off the furnace walls. Because of this effect, some of the mass 39 observations at low ion intensities were rejected in the analysis of the data. Also, in order to improve the signal-to-noise ratio at lower ion intensities, an ionizing energy of 20 volts was used throughout the measurements.

Crystalline $K_2Si_2O_5$ was prepared by the solid state reaction at 800 °C of Brazilian Optical Quality SiO_2 (purity > 99.99 percent as determined by NBS spectrochemical analysis) and anhydrous K_2CO_3 in weighed amounts corresponding to the appropriate composition. Repeated grinding and firing gave a product whose x-ray diffraction pattern showed only $K_2Si_2O_5$.

Preliminary vapor pressure measurements [7] were carried out by a gravimetric effusion method employing a Mettler Microbalance System using a platinum effusion cell with a nominal 1 mm diam orifice. Additional unpublished preliminary measurements were made using a modulated beam Bendix time-of-flight mass spectrometer in which the effusion cell employed a 1/2 mm diam effusion orifice (B5 experiments). The latter measurements yielded pressures significantly higher than the former, and the partial molal heat for formation of $2K(g) + 1/2 O_2(g)$ from the solution was significantly less than for the microbalance measurements. This suggested that the microbalance measurements were in error because a temperature dependent rate process (probably diffusion through the melt)

governed the apparent pressure. As a further check, mass spectrometer measurements using the Bendix system were made on a sample contained in a cell with a 1 mm diam orifice (B1 experiments). These data were consistent with the microbalance data in that the rate of vaporization decreased more rapidly with composition and the heat of vaporization was greater than for the measurements employing a 1/2 mm diam orifice. In the preliminary microbalance measurements, it was concluded on the basis of currently available thermodynamic data that the K_2O-SiO_2 system would be binary and that only K and O_2 would be important vaporization products. This was shown to be the case in the B5 and B1 measurements by the observation that the ratio of O_2/K pressures as indicated by the O_2^+/K^+ ion current ratio was constant with temperature and composition.

4. Results

Absolute temperatures, the bulk K_2O weight percent in the melt, and the potassium pressures² are listed in table 2 in the order taken. Low temperature points rejected from consideration are not tabulated. All of the data in series 1214 were obtained at temperatures and compositions at which a crystalline silica phase and a silica saturated solution would exist if thermodynamic equilibrium were established. Also, some data points in series 1123 through 1129 belonging in the two phase region are identified with an a. The data were broken up into eight series for analysis in an attempt to determine if the partial molal heat of evaporation was composition dependent. Data were grouped into different series if changes had been made in the experimental arrangement or a different mass spectrometer constant was used. Some of the data were arbitrarily divided if it appeared that sufficient change in the composition of the solution had taken place to allow a reasonable estimate of the dependence of pressure on composition to be made.

The data from each series were fit by least squares to the empirical equation,

$$\log P_K = B_0 + B_1/T + B_2 \cdot N^2, \quad (6)$$

where B_0 , B_1 , and B_2 are constants and N is the mole fraction of K_2O that would exist if the sample were a homogeneous solution. Table 3 lists the series identification, the number of data points, the range of K_2O bulk composition, the values of the above constants and their standard deviations, and the standard deviation in $\log P_K$. The value of B_1 , which is related to the partial molal heat of evaporation of K_2O from the solution, appears to be constant within a reasonable error limit for all the data through series 1129, while the constant B_2 which accounts for the variation in pressure with composition averages out to be about ten excepting the deviant values for series 1122 and 1129. The standard deviation in B_2 in both these series is higher than in the other series, possibly because of too narrow a composition range in series 1122 and phase separation in series 1129. It was found that all the data through series 1129 could be fit to a single master equation with only a modest increase in the standard deviation of $\log P_K$. However, the series 1214 data does

not fit the master equation, the constant B2 indicating that the variation in pressure with composition becomes greater in this composition range. In addition, the decrease in the B1 constant might also be indicative of separation of the solution into two phases.

The data for each of the series are plotted in figures 3a through 3h. Each plot shows the composition range for points of varying designations. On each data point, the direction of temperature change between points is noted. An upward tic on a data point shows that the temperature was being increased during the run while a downward tic indicates that temperatures were being decreased. Each figure shows two lines which represent the pressures calculated from the master equation for the higher and lower K_2O bulk composition range shown on the graph with the exception of figure 3h. For the 1214 series, the two lines represent the higher and lower K_2O bulk compositions calculated from the least squares fit for the 1214 data only. In figures 3a through 3g, the fit of the data to the master equation appears to be excellent. The major deviation from the equation appears in series 1122 (fig. 3d) in which the data clearly deviate from the upper line. A redetermination of the mass spectrometer constant was made at this juncture in the data and this could account for the deviation. However, the deviation is only about ten percent.

According to the phase equilibrium diagram accepted for the K_2O-SiO_2 system [8], the concentration of K_2O in the silica saturated solution varies from 23 to 5 wt percent K_2O as the temperature increases from 1273 to 1873 K. As noted above, some of the data in series 1123 through 1129 were taken at temperatures and compositions which would lead to precipitation of cristobalite or tridymite and the formation of a silica saturated solution richer in K_2O than the composition corresponding to the bulk K_2O composition. All of the data in the 1214 series correspond to compositions and temperatures which would place the melt composition in the two phase region. If equilibrium were established in the two-phase region at an appropriate rate, it might be observed in the mass spectrometer measurements because of shifts in the composition of the saturated solution. For example, an increase in temperature would lead to an initial increase in the pressure which would slowly decrease as silica dissolved in the melt until equilibrium was reattained. Conversely, if the temperature were decreased, after an initial drop, the K pressure would slowly increase as silica precipitated from the melt in order to re-establish equilibrium. Attempts to observe these effects along the liquidus curve were not successful. It should be noted that even though the pressure of K still depends on the liquidus composition, the K pressure is uniquely determined by the temperature alone in this region of the phase diagram under equilibrium conditions.

The observation that the K pressure continued to fall with decrease in bulk composition of K_2O in series 1214 is of itself evidence that equilibrium was not established. Failure to achieve equilibrium could be due to unsaturation effects in the Knudsen cell

²¹ 1 atmosphere = 101325 pascals.

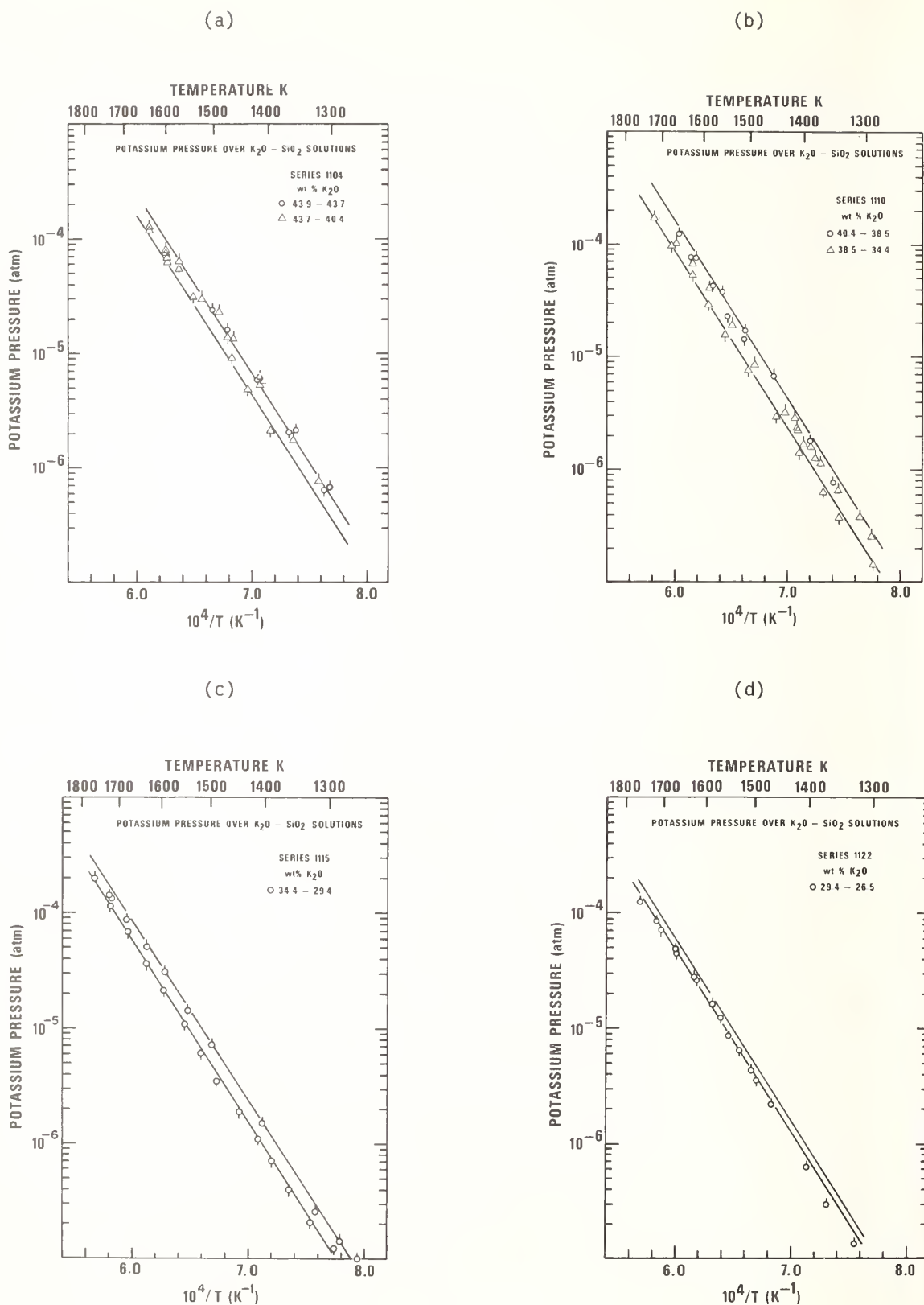
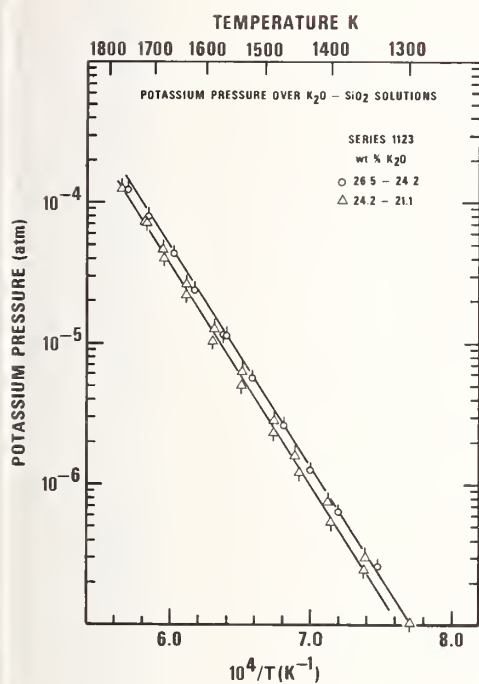
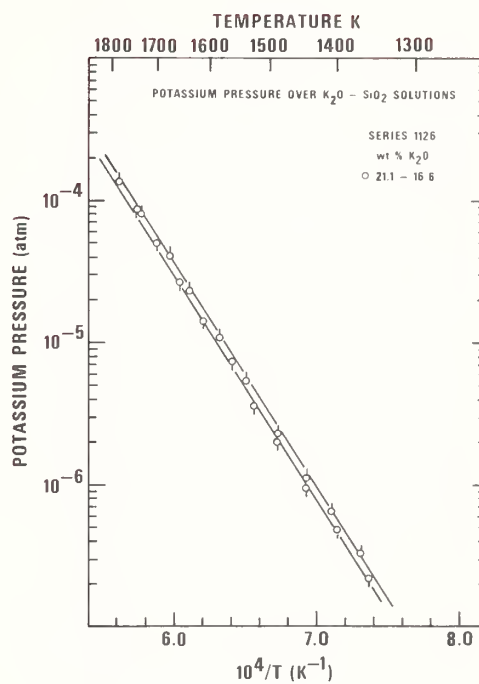


Figure 3. $\log P_K$ vs $1/T$ plots with lines representing pressures at upper and lower compositions based on master equation (a-g) or individual fit of data (h).

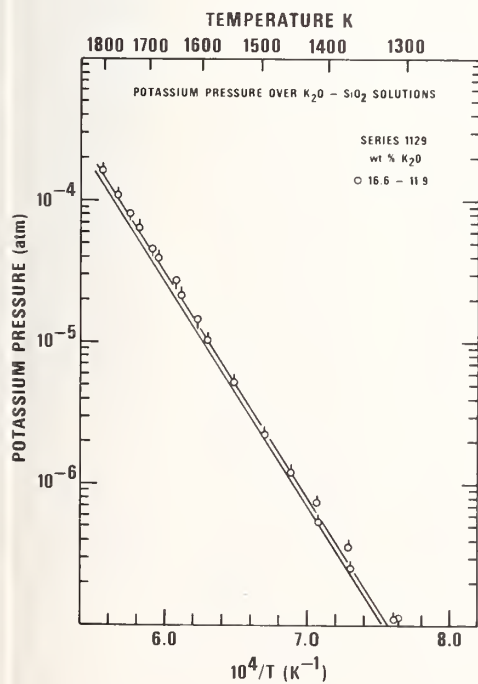
(e)



(f)



(g)



(h)

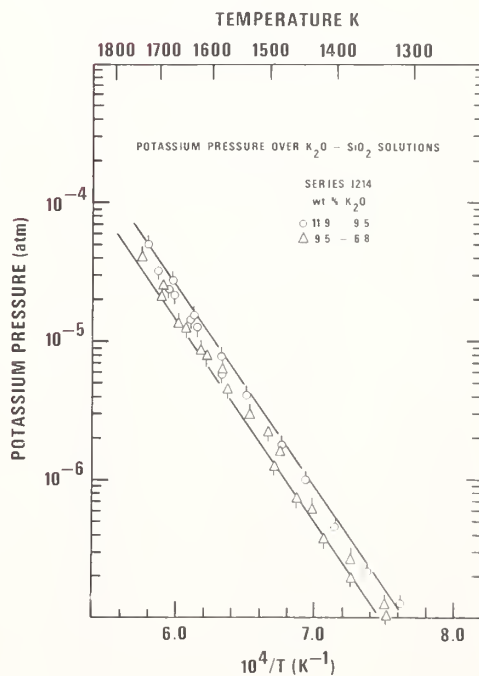


Figure 3. Continued

Table 2. Compilation of Data.

Series 1104			T,K	Wt % K ₂ O	P _K (atm)
T,K	Wt % K ₂ O	P _K (atm)			
			1489	38.31	8.79E-6
			1537	38.21	1.95E-5
1302	43.94	6.91E-7	1585	37.93	4.14E-5
1356	43.93	2.18E-6	1625	37.52	6.71E-5
1414	43.92	6.30E-6	1663	36.94	1.05E-4
1475	43.84	1.59E-5	1720	36.00	1.77E-4
1504	43.76	2.41E-5	1674	35.23	9.88E-5
1419	43.72	5.99E-6	1624	34.89	5.53E-5
1366	43.70	2.04E-6	1585	34.66	2.95E-5
1311	43.70	6.44E-7	1550	34.50	1.62E-5
1320	43.70	7.77E-7	1503	34.42	7.66E-6
1359	43.68	1.81E-6	1450	34.39	3.04E-6
1414	43.64	5.44E-6	1408	34.38	1.43E-6
1462	43.59	1.38E-5	1366	34.36	6.40E-7
1493	43.48	2.28E-5	1343	34.36	3.88E-7
1407	43.33	5.75E-6	1289	34.36	1.43E-7
1473	43.20	1.39E-5			
1524	43.04	3.02E-5		Series 1115	
1571	42.93	6.44E-5			
1571	42.51	5.63E-5	1259	34.36	1.00E-7
1601	42.51	8.22E-5	1283	34.36	1.43E-7
1601	42.02	7.73E-5	1320	34.35	2.59E-7
1640	42.02	1.29E-4	1404	34.34	1.53E-6
1640	41.32	1.22E-4	1435	34.34	2.27E-6
1594	41.32	6.91E-5	1496	34.30	7.28E-6
1594	40.99	6.32E-5	1544	34.24	1.43E-5
1542	40.73	3.16E-5	1592	34.03	3.12E-5
1466	40.44	9.21E-6	1631	33.74	5.17E-5
1436	40.39	4.91E-6	1681	33.40	8.92E-5
1396	40.36	2.16E-6	1723	32.81	1.42E-4
			1719	32.03	1.32E-4
			1765	31.18	2.03E-4
			1722	30.51	1.14E-4
1352	40.34	7.58E-7	1677	30.10	6.93E-5
1389	40.34	1.80E-6	1633	29.82	3.66E-5
1456	40.31	6.87E-6	1595	29.66	2.17E-5
1508	40.24	1.74E-5	1549	29.54	1.10E-5
1560	40.10	3.77E-5	1517	29.45	6.28E-6
1615	39.78	7.62E-5	1482	29.40	3.52E-6
1657	39.22	1.22E-4	1445	29.38	1.90E-6
1630	38.77	7.69E-5	1411	29.36	1.10E-6
1580	38.62	4.26E-5	1388	29.36	7.16E-7
1547	38.52	2.30E-5	1361	29.36	4.05E-7
1513	38.46	1.47E-5	1328	29.35	2.09E-7
1308	38.44	3.89E-7	1292	29.35	1.22E-7
1342	38.43	6.72E-7			
1369	38.43	1.17E-6		Series 1122	
1388	38.42	1.64E-6			
1416	38.42	2.90E-6	1294	29.34	9.48E-8
1292	38.41	2.65E-7	1325	29.34	1.36E-7
1413	38.40	2.43E-6	1369	29.34	3.00E-7
1379	38.38	1.34E-6			
1400	38.36	1.74E-6			
1411	38.35	2.23E-6			
1432	38.34	3.33E-6			

Table 2. (Continued)

T, K	Wt % K ₂ O	P _K (atm)	T, K	Wt % K ₂ O	P _K (atm)
1403	29.34	6.31E-7	1443	21.10	1.11E-6
1465	29.33	2.20E-6	1487	21.08	2.31E-6
1506	29.30	4.36E-6	1538	21.04	5.44E-6
1550	29.24	8.80E-6	1583	20.98	1.10E-5
1585	29.12	1.63E-5	1637	20.83	2.36E-5
1626	28.94	2.74E-5	1676	20.58	4.12E-5
1669	28.65	4.83E-5	1732	20.00	8.11E-5
1717	28.26	8.64E-5	1783	19.19	1.37E-4
1760	27.74	1.26E-4	1743	18.44	8.67E-5
1705	27.22	7.08E-5	1702	17.95	5.01E-5
1666	26.88	4.63E-5	1656	17.56	2.69E-5
1621	26.68	2.67E-5	1613	17.28	1.42E-5
1568	26.59	1.26E-5	1656	17.05	2.64E-5
1527	26.54	6.53E-6	1612	16.83	1.39E-5
1496	26.52	3.64E-6	1561 ^a	16.72	6.41E-6
Series 1123			1524 ^a	16.67	3.57E-6
1337	26.50	2.62E-7	1488 ^a	16.64	2.01E-6
1388	26.50	6.35E-7	1443 ^a	16.62	9.50E-7
1428	26.48	1.28E-6	1400 ^a	16.62	4.97E-7
1469	24.46	2.64E-6	1359 ^a	16.62	2.20E-7
1521	26.40	5.71E-6	1310 ^a	16.62	9.59E-8
1562	26.33	1.13E-5	Series 1129		
1620	26.18	2.45E-5	1373 ^a	16.60	3.64E-7
1660	25.92	4.37E-5	1416 ^a	16.60	7.52E-7
1713	25.48	7.92E-5	1308 ^a	16.59	1.15E-7
1758	24.74	1.26E-4	1313 ^a	16.59	1.11E-7
1568	24.24	1.15E-5	1316 ^a	16.59	1.11E-7
1352	24.20	3.22E-7	1368 ^a	16.59	2.60E-7
1403	24.20	7.70E-7	1410 ^a	16.58	5.71E-7
1451	24.18	1.67E-7	1454 ^a	16.58	1.22E-6
1484	24.17	2.91E-6	1494 ^a	16.56	2.27E-6
1535	24.14	6.50E-6	1543 ^a	16.52	5.36E-6
1584	24.06	1.30E-5	1589	16.42	1.05E-5
1634	23.90	2.64E-5	1636	16.25	2.18E-5
1681	23.58	4.68E-5	1681	15.94	3.90E-5
1722	23.16	7.56E-5	1723	15.50	6.37E-5
1770	22.55	1.30E-4	1767	14.88	1.10E-4
1717	21.96	7.14E-5	1800	13.93	1.65E-4
1679	21.61	4.08E-5	1741	13.02	8.16E-5
1634	21.38	2.25E-5	1693 ^a	12.47	4.59E-5
1587	21.24	1.07E-5	1646 ^a	12.10	2.74E-5
1537	21.18	5.14E-6	1608 ^a	11.92	1.47E-5
1484	21.16	2.40E-6	Series 1214		
1446	21.14	1.24E-6	1315 ^a	11.86	1.29E-7
1398	21.14	5.42E-7	1295 ^a	11.86	8.18E-8
1356 ^a	21.14	2.51E-7	1357 ^a	11.85	2.14E-7
1298 ^a	21.14	1.04E-7	1401 ^a	11.84	4.58E-7
Series 1126			1444 ^a	11.84	1.00E-6
1368 ^a	21.13	3.33E-7	1482 ^a	11.82	1.79E-6
1409	21.12	6.51E-7	1538 ^a	11.77	4.11E-6
			1582 ^a	11.68	7.81E-6
			1635 ^a	11.52	1.55E-5
			1678 ^a	11.21	2.70E-5
			1727 ^a	10.88	4.79E-5
			1707 ^a	10.64	3.21E-5
			1628 ^a	10.40	1.26E-5

^aCompositions and temperatures corresponding to 2-phase region.

Table 2. (Continued)

T, K	Wt % K ₂ O	P _K (atm)
1672 ^a	10.16	2.11E-5
1640 ^a	9.96	1.47E-5
1685 ^a	9.70	2.42E-5
1582 ^a	9.46	5.77E-6
1335 ^a	9.40	1.22E-7
1379 ^a	9.39	2.60E-7
1435 ^a	9.38	6.12E-7
1484 ^a	9.36	1.61E-6
1534 ^a	9.32	2.88E-6
1582 ^a	9.30	6.14E-6
1620 ^a	9.10	8.53E-6
1664 ^a	8.88	1.37E-5
1702 ^a	8.77	2.16E-5
1738 ^a	7.88	4.00E-5
1696 ^a	7.36	2.39E-5
1649 ^a	7.08	1.32E-5
1612 ^a	6.96	7.62E-6
1571 ^a	6.88	4.28E-6
1524 ^a	6.83	2.16E-6
1493 ^a	6.80	1.26E-6
1457 ^a	6.78	7.11E-7
1416 ^a	6.77	3.77E-7
1379 ^a	6.77	1.98E-7
1335 ^a	6.76	1.01E-7

Table 3. Least squares fit of K(g) pressures over K₂O-SiO₂ (ℓ) solutions.

Series	No. Pts.	K ₂ O wt. %	B0	S.D.	B1	S.D.	B2	S.D.	S.D.
1104	28	43.9-40.4	4.482	.131	-15637	192	12.02	1.29	.041
1110	38	40.3-34.4	4.790	.092	-15843	115	10.17	.66	.038
1115	26	34.4-29.4	4.559	.076	-15485	98	10.42	.83	.034
1122	18	29.3-26.5	5.349	.141	-15958	218	- 2.89	3.53	.047
1123	31	26.5-21.1	4.491	.067	-15305	93	10.35	1.05	.029
1126	23	21.1-16.6	4.784	.081	-15706	113	8.65	1.76	.030
1129	20	16.6-11.9	4.735	.092	-15226	182	- 9.77	5.46	.042
1214	37	11.9- 6.8	3.841	.076	-14654	116	52.76	4.27	.041
1104-1129	184	43.9-11.9	4.721	.047	-15624	72	9.32	.13	.057

caused by too large a ratio of orifice/sample area which would lead to a diffusion controlled vaporization process and/or because the rate of crystallization of cristobalite or tridymite is too slow. However, the deviation from equilibrium conditions may not be too great because the lower slope observed in the 1214 data would be expected if equilibration along the liquidus curve actually took place.

At the conclusion of the vaporization measurements, the residues remaining in the effusion cells used in two of the preliminary experiments and this experiment were examined by x-ray diffraction and/or using the SEM. The residue from the B5 experiment had a K_2O bulk composition of 20.5 wt percent. Its x-ray diffraction pattern showed only a broad peak characteristic of a K_2O-SiO_2 glass. The residue from the B1 experiment had a K_2O bulk composition of 18.8 wt percent K_2O . As noted earlier, it was concluded that there was a significant deviation from equilibrium conditions in this experiment. The x-ray diffraction pattern of this residue showed the presence of the broad peak characteristic of the K_2O-SiO_2 glass and a moderately strong pattern of a crystalline phase which was identified as low cristobalite. The x-ray pattern of the residue from the current experiment, which had a K_2O bulk composition of 6.8 percent, showed a strong pattern characteristic of low cristobalite as well as one weak peak characteristic of tridymite.

All three effusion cells showed signs of some migration of the sample. The bulk of the sample was contained in the bottom of the effusion cell but many small globules had wet the platinum and were distributed over the lower half of the cell. The globules were either clear and glassy or had a slightly milky appearance. A number of these from the B1 experiment were examined in situ by a SEM, but all appeared the same under magnification. At high magnification some crystallinity was apparent. The EDX spectra of globules selected at random showed the presence of Si and K with the K intensity somewhat less than might be expected, but did not reveal any significant variation in composition of various globules.

Under the conditions where the phase diagram indicates that cristobalite or tridymite should crystallize from the melt, low cristobalite which is a metastable phase was observed. Although the composition of the melt in metastable equilibrium with low cristobalite is not well known, Kracek, Bowen, and Morey [9] indicate a melt composition several percent poorer in K_2O than for the stable equilibrium. It is doubtful, however, that metastable equilibrium throughout the two phase region was actually achieved because the vaporization behavior would be expected to be similar to that for thermodynamic equilibrium. Quite a bit of data taken in the two phase region, however, does appear to represent equilibrium pressures in that the data agree satisfactorily with data representing the homogeneous solution. Substitution of the saturated solution composition for the calculated bulk composition would have little effect on the fit of the data to the master equation because the pressure is fairly insensitive to composition as the phase boundary is approached. The series 1214 data, however, probably yields low K pressures simply because there is insufficient sample left in the effusion cell to yield equilibrium pressures.

As stated earlier, the potassium pressure measurements can be used to determine the activity of K_2O in the K_2O-SiO_2 solution. For some calculations, an activity defined with

respect to a reference state of K(g) and O₂(g) may be useful and an activity may be defined as:

$$A_{K_2O} \text{ (ref. to vapor)} = P_K^2 P_{O_2}^{1/2} . \quad (7)$$

For these experiments, $P_{O_2} = 0.226 P_K$ and substitution of this into the master equation yields the equation

$$\log A_{K_2O} \text{ (ref. to vapor)} = 11.482 - 39060/T + 23.30 N^2. \quad (8)$$

which represents the activity of K₂O in the solution from a wt. fraction of 0.44 to the saturated solution phase boundary referenced to the vapor state. For other calculations, K₂O [ℓ] will a more useful reference state and eq. (8) may be modified to

$$\log A_{K_2O} \text{ (ref to } K_2O(\ell)) = \log A_{K_2O} \text{ (ref. to vapor)} - \log K_{eq} \quad (9)$$

in which K_{eq} is the equilibrium constant for reaction (1) when K₂O is a pure liquid.

5. Discussion

The results obtained in this work in general have good internal as well as external consistency. For example, the mass spectrometer constants were quite reproducible and their constancy during the measurements is shown by the fact that data from various experiments show the proper trend with composition, i.e., decreasing pressure with decreasing K₂O content. The partial molal heat of evaporation of K₂O from the melt is constant within error and is 178.7 kcal mol⁻¹ which is quite consistent with the heat of solution of K₂O in SiO₂ to form K₂Si₂O₅(-ℓ) (-68 kcal/mol⁻¹) estimated from Kelley's [10] compilation and the heat of formation of K₂O of -109 kcal mol⁻¹ both at 1500 K. The heat of formation is based on the JANAF [11] K₂O (c) table to the melting point, the heat of fusion reported by Natola and Touzain [12] and an estimated C_p of 25 cal mol⁻¹K⁻¹ for liquid K₂O.

The measurements presented provide activity data for K₂O in SiO₂ solutions from approximately 44 wt percent K₂O to the phase boundary and at temperatures from about 1300-1800 K. Some rough estimates of the overall possible systematic errors in the measurements may be useful. The largest error is undoubtedly due to error in temperature measurement which probably does not exceed 20 K. The error in composition is estimated to be not in excess of two percent at the starting composition and five percent at the final composition. At 1500 K, the estimated 20 K error yields an error in K pressure of about 40 percent.

It would be of scientific interest to measure the K₂O activity starting from the K₂O·SiO₂ composition and to make additional careful measurements using still smaller effusion orifices in the saturated silica solution area in order to cover a broader range

and to attempt to achieve equilibrium measurements in this difficult region of the system. The measurements here, however, cover most of the range of importance to MHD research for this simple system.

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